

Map Prop Engine – Quick Reference

URL parameters

darkMode sets darkMode
scene defines where assets, config and points are coming from ("data/{scene}/...")
limitMovementToWorld inhibit moving out of the world space. overwrites this setting from config.json.

config.json Structure

Defines map geometry, zoom limits, and image layers per zoom level (LOD).

```
{
  "worldWidth": 10000,
  "worldHeight": 10000,
  "initialView": { "x": 2048, "y": 2048, "z": 0.25 },
  "marker": { "display": true, "x": 2200, "y": 1800, "angle": 45 },
  "minZoom": 0.05,
  "maxZoom": 4,
  "forceType": "Mallorca",
  "limitMovementToWorld": true,
  "scaleAtZoom1": 1.5, // = real-world units per pixel at z=1 (e.g.,
  metersPerPixelAtZ1).
  "lods": [
    {
      "maxZ": 0.3,
      "grid": { "size": 200, "color": "rgba(255,0,0,.5)" },
      "layers": [
        { "url": "sat_low.png", "type": "base", "width": 2028,
"height": 1488, "offsetX": 0, "offsetY": 0 },
        { "url": "sat_low.png", "type": "base", "width": 2028,
"height": 1488, "offsetX": 2028, "offsetY": 0 },
        { "url": "roads_low.svg", "type": "overlay", "width": 1600,
"height": 900, "offsetX": 0, "offsetY": 0 }
      ]
    },
    {
      "maxZ": 0.9,
      "grid": { "size": 100, "color": "rgba(0,255,0,.5)" },
      "layers": [
        { "url": "sat_mid.png", "type": "base", "width": 2448,
"height": 1530, "offsetX": 0, "offsetY": 0 },
        { "url": "roads_mid.svg", "type": "overlay", "width": 1600,
"height": 900, "offsetX": 500, "offsetY": 0 }
      ]
    }
  ]
}
```

```

    ]
  },
  {
    "maxZ": 99,
    "grid": { "size": 50, "color": "rgba(0,128,255,.5)" },
    "layers": [
      { "url": "sat_hi.png", "type": "base", "width": 2640,
"height": 1588, "offsetX": 0, "offsetY": 0 },
      { "url": "roads_hi.svg", "type": "overlay", "width": 1600,
"height": 900, "offsetX": 0, "offsetY": 0 }
    ]
  }
],
"fakeLayers": [
  {"type": "Military Secrets"},
  {"type": "Streetmap"}
],
"presets": [
  { "label": "HQ Gate", "x": 1800, "y": 1800, "z": 0.5 },
  { "label": "Warehouse", "x": 3100, "y": 2100, "z": 0.8 },
  { "label": "Main Sq.", "x": 2048, "y": 2048, "z": 1.2 }
]
}

```

Scaling map images

Width & height needs the same proportions as the imgs, but no necessarily be the dimensions of the img.

So an image can be 1000 x 1000px wide but be defined as 666 x 666px to scale it down.

Use the bolt-button and the debug button to see the images all at once.

Find the correct X/Y position and offset of these lods by using the dev tools of your browser.

Adjust for left/right and fill those as offsets X/Y.

Find POIs

if in debug mode, right click on the world to see X/Y/Z of that point.

This also copies a valid json object to your clipboard for isnering into `points.json`.

Fields

Key	Description
<code>worldWidth</code> , <code>worldHeight</code>	total world pixel size (sets coordinate system)
<code>initialView</code>	starting camera position
<code>marker</code>	marker position (stays constant size)
<code>minZoom</code> , <code>maxZoom</code>	zoom limits

Key	Description
<code>forceType</code>	If not empty, this is the text that gets force-typed into the search bar. On enter, you fly to saved preset nr 1 (or to initialView if no presets defined).
<code>limitMovementToWorld</code>	Don't go out of bounds (outside worldWidth/height)
<code>scaleAtZoom1</code>	meters per pixel at zoom = 1 (for scale bar)
<code>lods</code>	Array of objects with data, layer image sets per zoom range etc
<code>fakeLayers</code>	additional non-working layers displayed in the layers list
<code>presets</code>	optional named camera positions (fly-to these by pressing 1...x on keyboard)

Each object in `config.lods` has these keys:

- `maxZ`: Max zoom where this lod is visible. Below this, its visible; above it is not.
- `grid`: Spacing and color of the grid. Make empty String if you don't want any grid showing.
- `layers`: Array of objects of individual image/SVG files to draw. Use multiple to tile or stack iamges (for instance , streets over satellite imagery).

Each object in `config.lods.layers` has these keys:

- `url`: path to image or SVG
- `type`: category that shows up in layer list (used for toggling on/off)
- `width` / `height`: native pixel dimensions
- `offsetX` / `offsetY`: optional placement offset

points.json (POIs)

Defines interactive points of interest (icons, text, tooltips, actions).

Get this data copied to your pasteboard when right clicking while in debug mode.

```
[
  {
    "tooltip": "This is videocam 1234123-A",
    "icon": "videocam",
    "x": 666,
    "y": 333,
    "minZ": 0.4,
    "maxZ": 2.5,
    "size": 2,
    "iconClasses": "red",
    "pointClasses": "blue",
    "iconStyles": "",
    "text": "Lorem Ipsum",
    "action": ""
  }
]
```

```

},
{
  "tooltip": "This is videocam 1234123-B",
  "icon": "location_pin",
  "x": 999,
  "y": 333,
  "minZ": 0,
  "maxZ": 4,
  "size": 2,
  "iconClasses": "",
  "pointClasses": "red text-shadow--white noBox",
  "iconStyles": "",
  "text": "",
  "action": "alert('cam A - AAAAA')\"
}
]

```

Hint: Add class `noBox` to a poi so there's no box around it. The bottom center of the icon is then at the defined point.

Key	Meaning
<code>x, y</code>	world coordinates
<code>icon</code>	emoji or short text
<code>size</code>	relative size (1 = 10 px base)
<code>minZ / maxZ</code>	visible zoom range
<code>pointClasses</code>	optional CSS styling for the whole bubble
<code>iconClasses / iconStyles</code>	optional CSS styling for the icon
<code>action</code>	JS snippet to execute on click

Grids

Each LOD can define its own grid overlay:

```
"grid": { "size": 100, "color": "rgba(255,255,255,0.2)" }
```

- **Size:** grid cell spacing in world pixels
- **Color:** any CSS color. if no color is set, it is not rendered for that LOD.